

Research Paper Summary Submission

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Research Paper Summary – Paper 1

Title: Natural Language Counterfactual Explanations in Financial Text Classification: A Comparison of Generators and Evaluation Metrics.

Authors: 1. Karol Dobiczek

2. Patrick Altmeyer

3. Cynthia C. S. Liem

Year: 2025

Publisher: Association for Computational Linguistics

Summary:

Problem Statement: The growing use of large language models (LLMs) in **high-stakes specialist fields like finance, particularly for classifying central bank monetary policy communications** (e.g., press releases as hawkish or dovish), necessitates high levels of trustworthiness and explainability. Counterfactual Explanations (CEs) offer a way to explain a model's output by proposing alternative inputs that would change its classification. However, existing CE generation methods and their evaluations often rely on generic tasks and datasets, utilizing **imprecise quantitative and qualitative metrics** that are frequently not rigorously defined, leading to inconsistent human assessments. This lack of precise evaluation hinders the development of truly reliable and plausible explanations in critical domains.

Methodology/Approach: The study evaluates three categories of CE generators: **LLM-assisted generation (Polyjuice), latent perturbation and decoding (PPLM), and sequential generation (RELITC)**. These generators were tested on a financial dataset from the Federal Open Market Committee (FOMC), consisting of central bank communications classified as hawkish, dovish, or neutral. The evaluation employed a dual approach: **quantitative metrics** (including perplexity, edit distance, embedding distance, faithfulness, and success rate) and **qualitative human evaluations**. Human evaluators included both non-expert native English speakers and central bank employees, who rated CEs for **fluency and plausibility**. The researchers refined definitions for these qualitative metrics to reduce ambiguity.

Results/Findings: The research found that **state-of-the-art text CE generators performed poorly on texts from specialist domains**. A significant finding was the **lack of alignment between most quantitative metrics and human evaluations** for fluency. For instance, RELITC, despite having poor perplexity scores, produced the most fluent texts according to human evaluators, while PPLM showed the opposite trend. Expert evaluators, however, gave **low plausibility scores to the generated counterfactuals**, noting that methods like RELITC and PPLM often misused domain-specific words, rendering explanations factually incorrect or unclear. Both non-experts and experts preferred CEs with minimal changes. The study also highlighted an inherent **trade-off between faithfulness and plausibility**.

Limitations/Future Work: The study's limitations include evaluating only three CE generation methods and relying on a **single financial dataset**, which lacks ground-truth counterfactuals. This means the findings might not generalize to other specialist domains or datasets. Future work should involve **evaluating text counterfactuals on non-trivial specialist tasks** from other domains like medicine or legal texts to develop a more generalized benchmark. The authors also recommend further exploration of involving the classifier in the CE generation process for improved faithfulness.

Personal Insight/Reflection: This paper strongly reinforces the crucial role of **human judgment, especially by domain experts**, in evaluating explainable AI, particularly in high-stakes contexts where nuances and domain-specific vocabulary are paramount. The significant discrepancy between quantitative metrics and human perception of qualities like plausibility underscores that purely algorithmic evaluation is insufficient for assessing the true utility and interpretability of CEs. It highlights that transparency in AI is not just about model mechanisms but also about the human comprehension and trust enabled by the explanations provided.

Research Paper Summary – Paper 2

Title: S-CFE: Simple Counterfactual Explanations

Authors: 1. Shpresim Sadiku

2. Moritz Wagner

3. Sai Ganesh Nagarajan

4. Sebastian Pokutta

Year: 2025

Publisher: Shpresim Sadiku

Summary:

Problem Statement: Machine learning models, particularly in critical decision-making domains, require transparency, fairness, and accountability. **Counterfactual Explanations (CFEs)** offer "what-if" scenarios to provide insight into a classifier's decisions by suggesting minimal input changes that alter the outcome. High-quality CFEs should adhere to five core principles: **Proximity** (small changes), **Validity** (desired outcome), **Actionability** (realistic changes), **Plausibility** (alignment with target class data distribution), and **Sparsity** (modifying as few features as possible). However, existing CFE methods frequently struggle to simultaneously achieve both sparsity and plausibility, often resulting in unrealistic or impractical explanations. This is complicated by the non-convex and non-smooth nature of classifier loss functions, manifold alignment metrics, and sparsity terms (like ℓ_0 regularizers) in the canonical CFE optimization problem.

Methodology/Approach/Case Studies: The paper introduces **S-CFE (Simple Counterfactual Explanations)**, a novel framework to generate optimal sparse and manifold-aligned CFEs by tackling the canonical optimization problem. S-CFE employs the **Accelerated Proximal Gradient (APG) method**, an efficient first-order procedure capable of handling smooth non-convex objectives and non-smooth ℓ_p regularizers (where $0 \leq p < 1$), which is crucial for enforcing true sparsity. The framework seamlessly integrates various classifiers and **differentiable plausibility measures**, such as Kernel Density Estimation (KDE), Gaussian Mixture Models (GMM), or a k-Nearest Neighbors-based density gravity approximation (S-CFEKDE, S-CFEGMM, S-

CFEkNN). Crucially, it controls the number of altered features precisely by using an indicator function for the sparsity constraint. Experiments were conducted on real-world datasets including **Boston Housing, Breast Cancer Wisconsin, Wine, and MNIST**, utilizing both 2-layer ReLU networks and Convolutional Neural Networks (CNNs) as classifiers.

Results/Findings: S-CFE consistently **outperforms existing benchmarks** (DCFE, PCFE, CEM) across datasets and classifiers, particularly in terms of **sparsity and adherence to the data manifold (lowest Local Outlier Factor or LOF)**. It achieves near-perfect **validity** for generated CFEs and comparable or slightly better **proximity** to factual data, all while maintaining computational efficiency. The integration of differentiable plausibility terms effectively guides the search towards high-density regions. Furthermore, incorporating plausibility constraints was shown to **improve robustness** against input manipulations, enhancing individual fairness of CFEs.

Limitations/Future Work: A primary limitation highlighted is that while sparse plausible CFEs indicate which feature changes *could* alter predictions, they **offer no guidance on real-world interventions** for achieving the desired outcome, as this requires causal knowledge. S-CFE, in its current form, guides users toward changes for predictor acceptance rather than improving the actual real-world state (e.g., changing symptoms might alter a diagnosis but not the actual infection status). Future work is suggested to **explore training S-CFE directly on data** rather than solely relying on classifier predictions.

Personal Insight/Reflection: This paper makes a significant stride in addressing the practical limitations of CFE generation by providing a method that produces **both sparse and plausible explanations**. The emphasis on simultaneously optimizing these often-conflicting objectives, especially through the use of APG for non-convex and non-smooth problems, is highly valuable. However, the acknowledged limitation regarding the distinction between prediction alteration and real-world causal intervention is a profound reminder that even the most advanced interpretability tools like S-CFE are still primarily aids for *understanding* model behavior, not substitutes for a deeper causal understanding of the underlying phenomena they model. This underscores the ongoing challenge of aligning algorithmic explanations with human intuition and actionable insights in complex domains.

